8331D

Silver Conductive Epoxy Adhesive

8331D is an electrically conductive, silver-filled 2-part epoxy adhesive. It is smooth, non-sagging, thixotropic, and bonds well to a wide variety of substrates.

This product allows for quick, cold-soldering repairs. It can also be used as a solder replacement for bonding heat-sensitive electronic componenets, or for making conductive connections where soldering is not an option, such as when bonding to glass, soft metals, or plastics.

8331D has been formulated to be economical. For a higher fill that maximizes conductivity, use 8330D. For a longer working time, use 8331S.

Features & Benefits

Creates strong permanent electrical connections

Cures at room temperature

SVHC free

Excellent Adhesion to many substrates

Room temperature storage

Long shelf life

Cure Instructions

Allow to cure at room temperature for 6 hours, or cure the adhesive in an oven at one of these time/temperature options:

Temperature	65 °C	80 °C
Time	10 min	5 min

Available Packaging

Part #	Packaging	Net Vol.	Net Wt.
8331D-14G	2 Syringe kit	6 mL	14.4 g
8331D-120G	2 Jar kit	50 mL	120 g

Storage and Handling

Store between 16 and 27 ˚C in a dry area, away from sunlight (see SDS). To maximize shelf life, recap product firmly when not in use.

8331D

Liquid Properties

Density	2.4 g/mL (Mixed) 2.5 g/mL (A) 2.3 g/mL (B)	ASTM D1475
Viscosity @ 25 ˚C	130 Pa·s (A) 130 Pa·s (B)	Brookfield Engineering labs Inc. IPCTM-65- Method 2.4.24.4
Mix Ratio	1:1 (Volume) 1.1:1 (Weight)	—
Working Time [a]	35 min	—
Set Time [a]	40 min	—
Peak Exotherm [b]	28 °C	ASTM D2471
Shelf Life	3 y	—

[a] Ambient temperature of 22 °C
[b] Based on a 250 g sample in a fixed container geometry.

Cured Properties

Color	Silver grey	—
Density	2.5 g/mL	Hydrostatic Weighing
Service Temperature Range	-50–150 ˚C	—
Resistivity	1.8×10^{-3} Ω·cm	ASTM D257
Hardness	78 D	ASTM D2240
Tensile Strength	13 N/mm^2	ASTM D638
Compressive Strength	69 N/mm^2	ASTM D695
Lap Shear	5.6 N/mm^2 (Stainless steel) 5.1 N/mm^2 (Aluminum)	ASTM D1002
Glass Transition Temperature (T_g)	35 ˚C	ASTM E1545
Coefficient of Thermal Expansion (CTE)	58 ppm/˚C (Prior T_g) 234 ppm/˚C (After T_g)	ASTM E831
Thermal Conductivity @ 25 ˚C Specfic Heat Capacity @ 25 ˚C Thermal Diffusivity @ 25 ˚C	1.5 W/(m·K) 0.7 J/(g·K) 0.9 mm^2/s	ASTM E1461
Weight Loss @ 155 °C (600 hrs)	2.9 %	–

Application Instructions

Read the product SDS for more detailed instructions before using this product.

Recommended Preparation

Clean the substrate with Isopropyl Alcohol, MG #824, so the surface is free of oils, dust, and other residues.

Syringe

1. Twist and remove the cap from the syringe. Do not discard cap.
2. Measure 1 part by volume of A.
3. Measure 1 part by volume of B.
4. Dispense material on a mixing surface or container, and thoroughly mix parts A and B together.
5. To stop the flow, pull back on the plunger.
6. Clean nozzle to prevent contamination and material buildup.
7. Replace the cap on the syringe.

Can or Jar

1. Stir each part individually to re-incorporate material that may have separated.
2. Measure 1.1 part by weight of A.
3. Measure 1 part by weight of B.
4. Thoroughly mix parts A and B together.
5. Apply adhesive to the application area.

Disclaimer: This information is believed to be accurate. It is intended for professional end-users who have the skills required to evaluate and use the data properly. M.G. Chemicals Ltd. does not guarantee the accuracy of the data and assumes no liability in connection with damages incurred while using it.